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import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
from sklearn.linear_model import LinearRegression

np.random.seed(42)

X = np.linspace(0, 10, 200).reshape(-1, 1)
y = np.sin(X).ravel() + np.random.normal(0, 0.15, 200)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

linear_predictions = linear_model.predict(X_test)

linear_rmse = np.sqrt(
    mean_squared_error(y_test, linear_predictions)
)

class KNNRegressor:

    def __init__(self, k=5):
        self.k = k

    def fit(self, X, y):
        self.X_train = X
        self.y_train = y

    def predict(self, X):
        predictions = []

        for x in X:
            distances = np.sqrt(
                np.sum((self.X_train - x) ** 2, axis=1)
            )

            nearest_indices = np.argsort(distances)[:self.k]
            nearest_values = self.y_train[nearest_indices]

            predictions.append(np.mean(nearest_values))

        return np.array(predictions)

knn_model = KNNRegressor(k=5)
knn_model.fit(X_train, y_train)

knn_predictions = knn_model.predict(X_test)

knn_rmse = np.sqrt(
    mean_squared_error(y_test, knn_predictions)
)

print("Model Performance")
print("-----")
print(f"Linear Regression RMSE : {linear_rmse:.4f}")
print(f"kNN Regression RMSE   : {knn_rmse:.4f}")

if knn_rmse < linear_rmse:
    print("\nkNN Regression performs better.")
else:
    print("\nLinear Regression performs better.")

sort_index = np.argsort(X_test[:, 0])

X_sorted = X_test[sort_index]
linear_sorted = linear_predictions[sort_index]
knn_sorted = knn_predictions[sort_index]

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plt.figure(figsize=(10, 6))

plt.scatter(X_test, y_test, label="Actual Data")
plt.plot(X_sorted, linear_sorted, label="Linear Regression")
plt.plot(X_sorted, knn_sorted, label="kNN Regression")

plt.xlabel("X")
plt.ylabel("Y")
plt.title("kNN Regression vs Linear Regression")
plt.legend()
plt.grid(True)
plt.show()
```

Model Performance

Linear Regression RMSE : 0.6508

kNN Regression RMSE : 0.1454

kNN Regression performs better.

